

A Technical Introduction to Part of speech tagging

Section 1

The sailor dogs the hatch. Correct grammatical tagging will reflect that "dogs" is here used as a verb, not as the more common plural noun. Grammatical context is one way to determine this; semantic analysis can also be used to infer that "sailor" and "hatch" implicate "dogs" as 1) in the nautical context and 2) an action applied to the object "hatch" (in this context, "dogs" is a nautical term meaning "fastens (a watertight door) securely").

Section 2

In corpus linguistics, part-of-speech tagging (POS tagging, PoS tagging, or POST), also called grammatical tagging, is the process of marking up a word in a text (corpus) as corresponding to a particular part of speech, based on both its definition and its context. A simplified form of this is commonly taught to school-age children, in the identification of words as nouns, verbs, adjectives, adverbs, etc. Once performed by hand, POS tagging is now done in the context of computational linguistics, using algorithms which associate discrete terms, as well as hidden parts of speech, by a set of descriptive tags. POS-tagging algorithms fall into two distinctive groups: rule-based and stochastic. E. Brill's tagger, one of the first and most widely used English POS taggers, employs rule-based algorithms.

Section 3

Tag sets Schools commonly teach that there are 9 parts of speech in English: noun, verb, article, adjective, preposition, pronoun, adverb, conjunction, and interjection. However, there are clearly many more categories and sub-categories. For nouns, the plural, possessive, and singular forms can be distinguished. In many languages words are also marked for their "case" (role as subject, object, etc.), grammatical gender, and so on; while verbs are marked for tense, aspect, and other things. In some tagging systems, different inflections of the same root word will get different parts of speech, resulting in a large number of tags. For example, NN for singular common nouns, NNS for plural common nouns, NP for singular proper nouns (see the POS tags used in the Brown Corpus). Other tagging systems use a smaller number of tags and ignore fine differences or model them as features somewhat independent from part-of-speech. In part-of-speech tagging by computer, it is typical to distinguish from 50 to 150 separate parts of speech for English. Work on stochastic methods for tagging Koine Greek (DeRose 1990) has used over 1,000 parts of speech and found that about as many words were ambiguous in that language as in English. A morphosyntactic descriptor

in the case of morphologically rich languages is commonly expressed using very short mnemonics, such as Ncmsan for Category = Noun, Type = common, Gender = masculine, Number = singular, Case = accusative, Animate = no. The most popular "tag set" for POS tagging for American English is probably the Penn tag set, developed in the Penn Treebank project. It is largely similar to the earlier Brown Corpus and LOB Corpus tag sets, though much smaller. In Europe, tag sets from the Eagles Guidelines see wide use and include versions for multiple languages. POS tagging work has been done in a variety of languages, and the set of POS tags used varies greatly with language. Tags usually are designed to include overt morphological distinctions, although this leads to inconsistencies such as case-marking for pronouns but not nouns in English, and much larger cross-language differences. The tag sets for heavily inflected languages such as Greek and Latin can be very large; tagging words in agglutinative languages such as Inuit languages may be virtually impossible. At the other extreme, Petrov et al. have proposed a "universal" tag set, with 12 categories (for example, no subtypes of nouns, verbs, punctuation, and so on). Whether a very small set of very broad tags or a much larger set of more precise ones is preferable, depends on the purpose at hand. Automatic tagging is easier on smaller tag-sets.

Section 4

Use of hidden Markov models In the mid-1980s, researchers in Europe began to use hidden Markov models (HMMs) to disambiguate parts of speech, when working to tag the Lancaster-Oslo-Bergen Corpus of British English. HMMs involve counting cases (such as from the Brown Corpus) and making a table of the probabilities of certain sequences. For example, once you've seen an article such as 'the', perhaps the next word is a noun 40% of the time, an adjective 40%, and a number 20%. Knowing this, a program can decide that "can" in "the can" is far more likely to be a noun than a verb or a modal. The same method can, of course, be used to benefit from knowledge about the following words. More advanced ("higher-order") HMMs learn the probabilities not only of pairs but triples or even larger sequences. So, for example, if you've just seen a noun followed by a verb, the next item may be very likely a preposition, article, or noun, but much less likely another verb. When several ambiguous words occur together, the possibilities multiply. However, it is easy to enumerate every combination and to assign a relative probability to each one, by multiplying together the probabilities of each choice in turn. The combination with the highest probability is then chosen. The European group developed CLAWS, a tagging program that did exactly this and achieved accuracy in the 93–95% range. Eugene Charniak points out in *Statistical techniques for natural language parsing* (1997) that merely assigning the most common tag to each known word and the tag "proper noun" to all unknowns will

approach 90% accuracy because many words are unambiguous, and many others only rarely represent their less-common parts of speech. CLAWS pioneered the field of HMM-based part of speech tagging but was quite expensive since it enumerated all possibilities. It sometimes had to resort to backup methods when there were simply too many options (the Brown Corpus contains a case with 17 ambiguous words in a row, and there are words such as "still" that can represent as many as 7 distinct parts of speech. HMMs underlie the functioning of stochastic taggers and are used in various algorithms one of the most widely used being the bi-directional inference algorithm.